

Review article

Application of artificial intelligence in dental implant prognosis: A scoping review

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ABSTRACT

Objectives: The purpose of this scoping review was to evaluate the performance of artificial intelligence (AI) in the prognosis of dental implants.

Data: Studies that analyzed the performance of AI models in the prediction of implant prognosis based on medical records or radiographic images. Quality assessment was conducted using the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Quasi-Experimental Studies.

Sources: This scoping review included studies published in English up to October 2023 in MEDLINE/PubMed, Embase, Cochrane Library, and Scopus. A manual search was also performed.

Study selection: Of 892 studies, full-text analysis was conducted in 36 studies. Twelve studies met the inclusion criteria. Eight used deep learning models, 3 applied traditional machine learning algorithms, and 1 study combined both types. The performance was quantified using accuracy, sensitivity, specificity, precision, F1 score, and receiver operating characteristic area under curves (ROC AUC). The prognostic accuracy was analyzed and ranged from 70 % to 96.13 %.

Conclusions: AI is a promising tool in evaluating implant prognosis, but further enhancements are required. Additional radiographic and clinical data are needed to improve AI performance in implant prognosis.

Clinical significance: AI can predict the prognosis of dental implants based on radiographic images or medical records. As a result, clinicians can receive predicted implant prognosis with the assistance of AI before implant placement and make informed decisions.

1. Introduction

Dental implants have become one of the routine treatment methods for oral rehabilitation [1]. The long-term survival rate, which means the success rate of dental implants after 10 years, is over 96.4 % [2–5]. However, with the general increase of implants placed, the relative risk for complications that may lead to implant failure rises [6]. Factors affecting early and late failures differ and also depend on the timing of occlusal loading [7]. Risk factors that lead to the failure of dental

implants include but are not limited to smoking [8,9], untreated periodontal disease, poor bone quality, diabetes [10], low levels of vitamin D [11,12], hypercholesterolaemia [13] and lack of primary implant stability [14,15]. Additionally, mechanical overload may also cause bone resorption and implant fracture which result in implant failure [16]. Since implant failure is an end-result condition that requires explantation, identifying the main risk factors and predicting implant failure is important for both patients and dentists [14,17]. Due to the various pathogenic factors and clinical features, currently it is difficult

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for clinicians to predict implant success or failure [14]. Although vast amounts of data on clinical features were extracted from the electronic dental records to assist in the prediction of implant success, the accuracy of prediction remained unsatisfactory [18]. Thus, new methods for predicting the prognosis of dental implants are required.

Artificial intelligence (AI) was first defined as the intelligence displayed by machines made by humans at the Dartmouth Conference in 1956 [19]. AI is similar to human intelligence in thinking and behavior. AI can learn and partake in reasonable actions based on environmental conditions by simulating, extending, and expanding human intelligence [20]. Machine Learning (ML) is a branch of AI, which involves the development of algorithms that can learn based on input data and can be trained to provide predicted values. The algorithms are based on extracting patterns and selecting specific properties from a large amount of data while eliminating irrelevant information from the input sources [21]. Deep Learning (DL), based on artificial neural networks (ANN), is considered a hot topic because of its strong learning capabilities from the input data. Unlike standard ML models, DL technology represents the abstractions of data to build computational models with multiple layers, which takes a shorter time to test than other ML algorithms [22]. AI was widely applied to clinical medicine [23]. Advantages such as high efficiency and the ability to deal with complex conditions enable AI to cover the areas of clinical medicine [24] and dentistry [25]. In dentistry, AI is widely used in clinical diagnosis and treatment [25]. Different types of AI have emerged and have assisted in the diagnosis of cysts and tumors [26], detecting and classifying dental caries and peri-apical lesions [27–30], diagnosis of gingivitis and periodontitis [31], orthodontic treatment [25], pediatric dentistry [32], segmentation of nature teeth and maxillofacial structures [33–35], guide for dentists and dental researchers [36], and oral hygiene education for the public [37, 38].

In implant dentistry, AI has also been used in several aspects including the recognition and classification of different types of implants [39,40], segmentation of dental implants on radiographic images [41], preoperative planning on radiographic images [42,43], decision-making of whether to place a dental implant [44], and predicting implant prognosis after placement [45]. The first application of AI in the prediction of implant success was reported in 2005 [45], followed by more research using a variety of AI models. Several groups have reviewed the application of AI in implantology, to evaluate the performance of AI in dealing with different tasks related to implantation [46,47]. However, the applicability of AI models in the prediction of implant prognosis remains unclear. An assessment of the performance and limitations of the AI models for predicting the dental implant prognosis is recommended to assist dentists to make clinical decisions.

The use of AI in predicting implant prognosis needs to be explored. Therefore, to the best of the authors' knowledge, this scoping review is the first review that aimed to evaluate the performance of AI algorithms in the prediction of implant prognosis (i.e., osseointegration, peri-implantitis, dental implant failure, and marginal bone level). The null hypothesis was that AI does not accurately predict implant prognosis.

2. Materials and methods

The scoping review was carried out in accordance with the guidelines of PRISMA-ScR guidelines [48]. The study protocol was registered on Open Science Framework on 12 February 2024.

2.1. Search strategy

The focus question was “What is the performance of artificial intelligence models in evaluating implant prognosis according to the medical records of patients or radiographic images?”.

The population, intervention, comparison, and outcome (PICO) question was defined as follows:

Population: dental implants required to be assessed after implant

surgery;

Intervention: all types of AI models;

Comparison: human intelligence;

Outcome: performance of AI in the prognosis of dental implants according to medical records or radiographic data.

According to the PICO criteria, the search strategy was developed.

Studies published in English no later than October 11, 2023 in four different databases including PubMed MEDLINE, EMBASE, Scopus, and The Cochrane Library were searched. The electronic search strategy was listed in Supplementary Table 1. A manual search was also conducted in the following Journals: *Journal of Dentistry*, *Journal of Clinical Periodontology*, *Journal of Dental Research*, *Clinical Oral Implants Research*, *Clinical Implant Dentistry and Related Research*, *The International Journal of Oral & Maxillofacial Implants*, and *Journal of Prosthetic Dentistry*. An attempt was also made to identify unpublished trials and gray literature by searching the U.S. National Library of Medicine (ClinicalTrials.gov) and opengrey.eu.

2.2. Eligibility criteria

2.2.1. Inclusion criteria

- Studies on dental implant prognosis and peri-implantitis prediction using all types of AI models
- Studies that applied AI models to assess the osseointegration after dental implant placement
- AI performance evaluated using clinical features and/or radiographic images (both 2D and 3D imaging data)

2.2.2. Exclusion criteria

- Review articles and meta-analyses
- AI-related animal and in vitro studies
- Studies without performance analysis of the models
- Studies that used AI models for other surgical implant procedures including cardiac and orthopedic implants
- Studies that developed AI models for other fields in dentistry such as periodontology, oral and maxillofacial surgery, pediatric dentistry, orthodontics, tooth and bone segmentation, classification and recognition of dental implants, and pre-operative planning before implant surgery

2.3. Study selection and data extraction

Title and abstract screening was conducted by two reviewers (Z.W. and X.Y.) after removal of duplicates. Full texts of eligible studies were subsequently evaluated based on the inclusion and exclusion criteria. Any disagreements were resolved through discussion with two senior reviewers (F.W. and C.X.). Data from the included studies were extracted independently by the two reviewers. In each study, the first author, year of publication, study design, prospect, AI model, dataset size and distributions, parameters, performance, and comparison with dentists or other AI models were extracted.

2.4. Risk of bias assessment

The quality of the data was evaluated by applying the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Quasi-Experimental Studies. The checklist requires reviewers to choose between “Yes”, “No”, “Unclear” and “Not applicable” for 9 questions. High risk of bias was considered when a study tallied up to 49 % “yes” answers, moderate when a study tallied from 50 % to 69 % “yes” answers, and a low risk of bias when a study tallied more than 70 % “yes” answers. Any disagreements between the two reviewers were resolved through discussion with two senior investigators (F.W. and C.X.).

2.5. Data synthesis

AI performance was evaluated using a combination of accuracy, sensitivity, specificity, precision, F1 score, and receiver operating characteristic area under curves (ROC AUC). Due to the heterogeneity of the data, a meta-analysis could not be performed.

3. Results

3.1. Search strategy

The PRISMA flowchart is shown in Fig. 1. According to the search strategy detailed in Supplementary Table 1, the initial search resulted in 892 studies. After excluding 160 duplicates and 696 articles based on titles and abstracts, 36 full texts were screened. The excluded articles with reasons are shown in Supplementary Table 2. 12 articles published between 2017 and 2023 were included in the final analysis. There was significant agreement between the two reviewers during the study selection, with a Cohen Kappa of 0.82.

3.2. Risk-of-bias assessment

Quality assessment is shown in Supplementary Table 3. Questions 3, 5, and 7 of the JBI Critical Appraisal Checklist for Quasi-Experimental Studies were not applicable for any of the included studies. According to the standard, 6 studies recorded between 50 %–69 % and had a moderate risk of bias [49–54], while the other 6 tallied less than 49 %, which showed a high risk of bias [55–60].

3.3. Performance of the ai models

Detailed descriptions of the included studies are displayed in Table 1. Performance measured with accuracy, sensitivity, specificity, precision, F1 score, and ROC AUC are defined as follows:

$$accuracy = \frac{TP + TN}{TP + FP + TN + FN}$$

$$sensitivity / recall = \frac{TP}{TP + FN}$$

$$specificity = \frac{TN}{FP + TN}$$

$$precision / positive predictive values = \frac{TP}{TP + FP}$$

$$F1\ score = \frac{2 \times precision \times recall}{precision + recall}$$

Confusion Matrix		
	Prediction +	Prediction –
Ground Truth +	True Positive (TP)	False Negative (FN)
Ground Truth –	False Positive (FP)	True Negative (TN)

Accuracy was analyzed in 7 studies [49,52,53,55–57,60]. The accuracy of the AI models in predicting implant prognosis ranged from 70 % to 96.13 %. Among the studies that applied deep learning algorithms, sensitivity and specificity was calculated [51–53,57,60] and ranged from 67 % to 95.65 % and 78 % to 95.65 %, respectively. Precision was

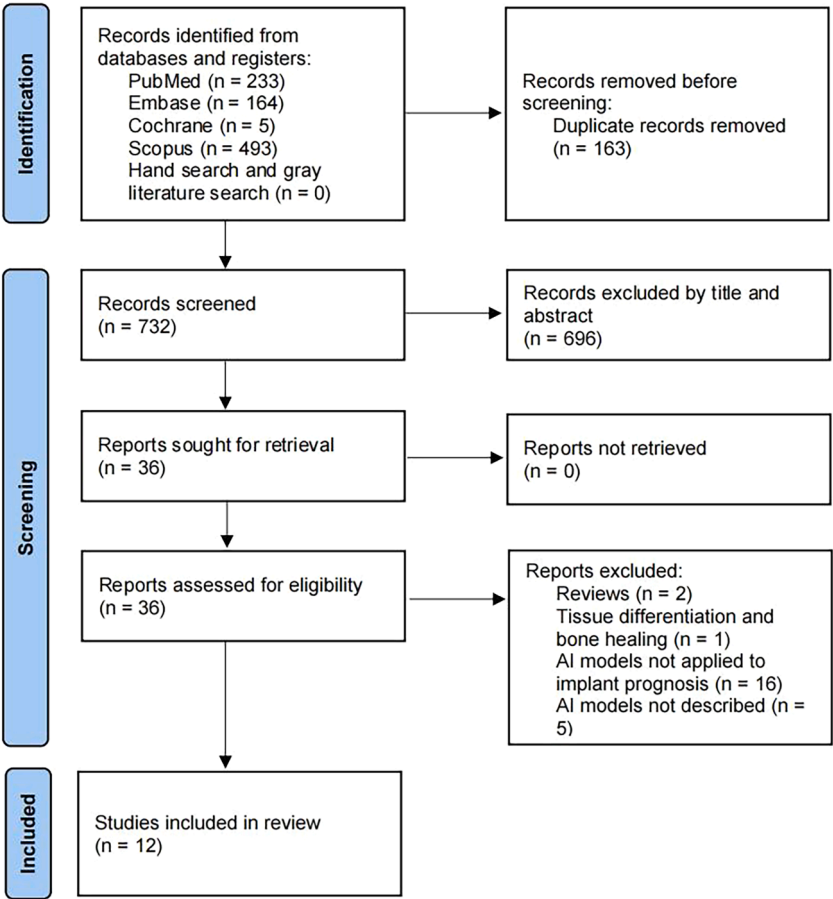


Fig. 1. PRISMA flowchart.

Table 1
Characteristics of the included studies.

Authors (year) Country	Study design	Prospect	AI model	Dataset size	Training dataset size	Validation dataset size	Testing dataset size	Features	Performance	Comparison
Cha JY et al. (2021) Korea	Retrospective study	Detecting the marginal bone level, top, and apex of implants	DL	708 periapical radiographic images	508 periapical radiographic images	100 periapical radiographic images	100 periapical radiographic images	Periapical radiographic images	Average position = 0.627 (upper) and 0.657 (lower); Average recall = 0.684 (upper) and 0.728 (lower)	No statistically significant difference was found between the modified R- CNN model and dental clinician for detecting landmarks around dental implants NR
Chen YC et al. (2023) China	Retrospective study	Evaluating the degree of periodontal damage around implants	DL	456 periapical films	147 periapical films	NR	46 periapical films	Periapical film	The accuracy rate was 90.4 % and the precision rate was 90.7 %	NR
Huang N et al. (2022) China	Retrospective study	Predicting the risk of dental implant loss	LR and DL	603 patients (279 high-risk and 324 low- risk); 680 dental implants (310 high- risk implants and 370 low-risk implants)	544 implants (Training&Validation dataset)	Fivefold cross validation	136 implants	CBCT; Clinical features	AUC=0.90, 95 % Confidence interval=0.84–0.95	Compared with 3 junior dentists and 3 senior dentists; the integrated model performed better NR
Huang Z et al. (2022) China	Retrospective study	Evaluating implant stability	DL	20,488 cross-sectional images of the arch and 779 implant images (CBCT)	624 implant images (CBCT)	NR	155 implant images (binary and ternary) and 150 implant images (quaternary); average time=3.76 s	CBCT	Binary, ternary,and quaternary implant stability quotient classification test set accuracies were 96.13 %, 95.33 %, and 92.90 %, with mean precisions of 96.20 %, 95.33 %, and 93.71 %	The deep learning model performed similarly to the resident dentist, but less well than the experienced dentist
Liu M et al. (2022) China	Retrospective study	Detecting marginal bone loss around implants	DL	1670 periapical radiographic images (835 with marginal bone loss around the implants and 835 without marginal bone loss)	1370 periapical radiographic images	150 periapical radiographic images	150 periapical radiographic images	Periapical radiographic images	Bone loss around implants and lesion sites, positive predictive values=81 % and 87 %, sensitivities=67 % and 75 %, specificities=87 % and 83 %	Comparing 6 developed architectures of neural networks and the neural network architecture NO.4 had the highest accuracy of 94.48 %
Lyakhov PA et al. (2022) Russia	Retrospective study	Predicting the survival of dental implants	DL	1626 cases of dental implantation (1490 cases of successful implantation and 136 cases of implant rejection)	1300 dental implants	NR	326 dental implants	55 clinical features	Highest accuracy=94.48 %; Highest F1-score=0.9657; best recall rate=0.9837; highest specificity index=0.9565	

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Table 1 (continued)

Authors (year) Country	Study design	Prospect	AI model	Dataset size	Training dataset size	Validation dataset size	Testing dataset size	Features	Performance	Comparison
Mameno T et al. (2021) Japan	Retrospective study	predicting the onset of peri- implantitis	LR, SVM, and RF	254 dental implants (127 peri-implantitis group and 127 non- peri-implantitis group)	178 dental implants	Fivefold cross- validations	76 dental implants	Clinical features	The highest performance for prediction of peri-implantitis (accuracy=0.70, precision=0.72, recall=0.66, f1-score=0.69).	RF had the highest classification performance among LR, SVM
Oh S et al. (2023) Korea	Retrospective study	Predicting the osseointegration for dental implant	DL	580 patients (338 osseointegration not yet and 242 successful osseointegration); 1206 dental implants (591 osseointegration not yet and 615 successful osseointegration)	347 patients	116 implants	117 implants	Panoramic and periapical radiographs	Best accuracy=0.896; Mean specificity=0.780–0.857, Mean sensitivity=0.811–0.833, Mean accuracy=0.799–0.836, Mean AUROC=0.890–0.922	Comparing 7 deep learning models, and ResNet-50 had the best accuracy
Papantonopoulos G et al. (2016) Greece	Retrospective study	Predicting the individual implant bone levels	K-means; principal component analysis; ensemble selection; support vector machine	72 patients (32 found peri-implantitis and 40 not found peri- implantitis); 237 dental implants (73 found peri-implantitis and 164 not found peri-implantitis)	NR	10-fold resampling cross- validation	NR	17 variables of clinical features	a cluster of implants susceptible to peri- implantitis and two overlapping clusters of implants resistant to peri- implantitis	NR
Rekawek P et al. (2023) United States	Retrospective study	Predicting implant failure and peri- implantitis	LR, RF, SVM, and Ensemble techniques	398 patients and 942 dental implants	754 dental implants	NR	188 dental implants	61 clinical features for implant failure; 89 clinical features for peri- implantitis	Random forest classifiers had the best AUCs, with the score of 0.870 for implant failure and 0.843 for peri- implantitis	Comparing 4 machine learning models and random forest classifiers performed the best
Vera M et al. (2023) Spain	Retrospective study	Detecting the peri- implant marginal bone	DL	2920 dental intraoral digital radiographs (350 bitewing radiographs and 1986 periapical radiographs, augmenting the initial dataset by 25 % for each type)	1314 dental intraoral digital radiographs	146 dental intraoral digital radiographs	1460 dental intraoral digital radiographs	Periapical and bitewing radiographs	Performance is satisfactory (without specific data)	NR
Zhang C et al. (2023) China	Retrospective study	Predicting the implant outcome	DL	185 patients (26 with failed implants and 159 with successful implants); 222 dental implants (63 failed and 159 successful); 900 panoramic radiographs and 900 periapical radiographs (augmented from the original 529 periapical images and 551 panoramic images)	720 panoramic radiographs and 720 periapical radiographs	Five-fold cross- validation	180 panoramic radiographs and 180 periapical radiographs	Periapical and panoramic images	Prediction accuracy=87 %, precision=0.85, recall=0.88, F1-score=0.85	The prediction accuracy is higher than the compared studies from other research fields

Abbreviations: **CBCT**, Cone-beam computed tomography; **DL**, Deep learning; **R-CNN**, Region Based Convolutional Neural Networks; **NR**, not reported; **LR**, Logistic regression; **SVM**, Support vector machines; **RF**, Random forest; **AUC**, area under the curve; **ROC**, Receiver operating characteristic.

measured in 5 studies [51,53,55,56,60], and ranged from 72 % to 96.2 %. F1 score is a measurement indicator that shows the difference between precision and recall, which was calculated in 3 articles [52,53,60] and ranged from 0.69 to 0.9657. ROC AUC shows the area under the receiver operating characteristic curves, which indicate the ability of the binary classification model to classify the samples in the dataset. Only 2 studies evaluated performance with AUC, which were 0.90 and 0.843, respectively [52,54].

3.4. Subgroup analysis: ai models

AI models were documented in all 12 articles (Fig. 2). Among these studies, 8 used deep learning for the prediction of dental implants, 3 applied traditional ML algorithms for postoperative implant prognosis, and 1 study combined both types together to realize the research objectives [49]. Features were extracted from the radiographic images and medical records of patients and trained in the DL model to predict the success or failure of implants depending on radiological and medical records. Conventional Neural Networks (CNN) are a common DL model, was utilized in 6 articles, and the accuracy ranged from 79.9 % to 96.13 %. Logistic regression, support vector machine, random forest, and ensemble selection were traditional ML models applied in 5 studies. Random forest had the best performance in classifying different groups of implant prognosis in 2 studies, with the highest accuracy of 0.70 and best AUC of 0.870, respectively.

3.5. Subgroup analysis: features of the samples in the dataset

Features of the samples were clarified in the 12 studies (Fig. 3). Seven extracted data from the radiographic images (e.g., panoramic, periapical, bitewing radiographs, and cone-beam computed tomography (CBCT)). The accuracy of predicting implant failure based on CBCT was the highest at 96.13 %. Features were mainly extracted from periapical and panoramic radiographs, and the test accuracy ranged from 79.9 % to 90.4 %. Four studies extracted features from the medical records of patients, implants, and surgeries. Medical records consisted of medical records, oral examination, and information on implants, crowns, and surgeries. The number of features varied from 17 to 89. The highest accuracy was 94.48 % while the lowest was 70 %. One study obtained features from both preoperative radiographic images and medical records, and the AUC of the study was 0.95.

4. Discussion

Difficulties and challenges exist for clinicians to predict implant prognosis after placement [5]. In this scoping review, the performance of all AI types in the prognosis of dental implants was analyzed and

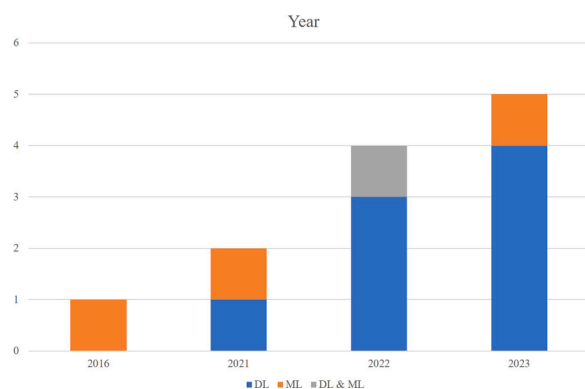


Fig. 2. Number of included studies using different types of artificial intelligence for implant prognosis.

Abbreviations: DL, Deep learning; ML, Machine learning.

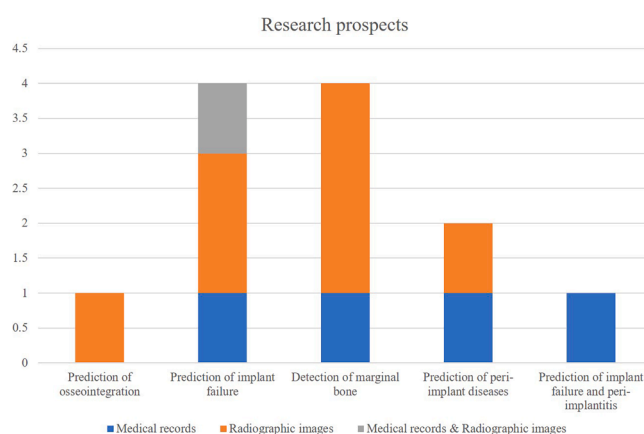


Fig. 3. Number of included studies using different sources of features for implant prognosis.

evaluated. The accuracy of AI models in predicting implant prognosis was relatively high ranging from 70 % to 96.13 %; therefore, the null hypothesis was rejected.

Notably, most studies included showed high accuracy, sensitivity, specificity, precision, F1 score, and AUC of AI models for predicting implant prognosis after placement, based on either radiographic images or clinical information. All AI models used were trained to extract features and predict results, assisting dentists to identify failed implants and make decisions for further treatment. Moreover, the indices to present the performance varied among the 12 studies. Three studies compared the performance of prediction between AI models and dentists. The results showed that the performance of AI is better than clinicians according to Huang et al., with no significant difference based on Cha et al., and similar to resident dentists but not as well as experienced dentists in the study conducted by Liu et al. It seemed that the effect of AI models to assist dentists in predicting implant prognosis differed between these 3 studies. Since the application scenarios, AI models, dataset size, and the features extracted are hardly the same, the result may be dependent on these influence factors. Therefore, dentists may decide whether to apply an AI model to help with the prognosis of implants based on specific clinical needs.

Eight studies trained AI models based on radiographic images [49–51,55–57,59,60] (1 study used both radiographs and medical records [50]) and all of them used DL models. Two of the 8 studies trained their models with features extracted from CBCT [50,56], and the performance is relatively better than that of the rest 6 which obtained features from radiographs including periapical films, panoramic radiographs and bitewing radiographs. Huang et al. showed the highest accuracy (96.13 %) and precision (96.20 %), which was at least 5 % higher than other models based on radiographic data. According to Polizzi et al. and Badr et al., AI models trained with CBCT tend to perform better in segmenting natural teeth and the anatomical structures in the oral cavity [61,62]. A single 3-dimensional radiograph is sliced into dozens if not hundreds of 2-dimensional images, increasing the number of samples for AI model training, and thus, improving the performance of the model. Features extracted from 3-dimensional images may be of superior quality than those extracted from 2-dimensional data, which may explain the similar implant prognostic ability though clear evidence regarding the superiority is currently not present. Periapical radiographs are taken right after placement as well as after full crown prosthesis as a routine method [63]. Six studies collected information from periapical films. The best performance is to predict the osseointegration for dental implants [57], while the worst is to detect the marginal bone level, top, and apex of implants [49]. Neto et al. reviewed that periapical radiographs are more suitable than panoramic radiographs in detecting the osseointegration of dental implants, and thus, assisting dentists in deciding the time for restoration [64]. Panoramic and bitewing

radiographs were used in 3 studies and compared with periapical radiographs [57,59,60], and the results showed that the precision was higher when trained with panoramic radiographs, but recall was lower than those trained with periapical films [60]. It indicated fewer false positive cases and more false negative cases when the AI models were trained with panoramic radiographs. Therefore, panoramic radiographs may be better used for prediction with high demand for decreasing the rate of missed detection of failed implants.

Five studies extracted features from medical records [50,52–54,58] (1 study combined radiographs with medical records [50]), and the AI included traditional ML models and DL models. The features for training consisted of general patient information, intraoral examination, dental implant indices, and surgical information. Age, smoking, implant length, implant diameter, implant torque, amount of local anesthetic, and oral hygiene were important factors for implant failure [50,52,54]. Implant length, implant diameter, oral hygiene, keratinized mucosa width, diabetes, age, and smoking influence the possibility of peri-implantitis [53, 54]. The accuracy of AI models trained by Lyakhov et al. surpassed that of other studies, possibly due to the largest training and test dataset sizes and the neural network applied in this study [52]. The dataset contained 1626 samples and 55 types of features were collected in order to elucidate the relationship between implant prognosis and risk factors which may explain the accuracy of 94.48 %. Random forest is a type of ML algorithm to classify samples based on input features [65]. It performed the best among the traditional ML models in predicting the implants since the task required the AI models to separate implants into 2 or more categories depending on the prognosis [53,54]. Unlike other studies which focused on classification problems, Papantonopoulos et al. aimed to cluster the patients and implants into 2 groups, susceptibility or resistance to peri-implantitis [58]. Different ML models were used in the study to clarify the internal relationship between the features. Fewer teeth, younger age, and jaw bone site were the three most important characteristics of the phenotypes, and the result was quite different from other research [66]. Despite the fast development of DL algorithms to replace ML models, the study could still identify the phenotypes of dental implants and provide a new thought in a certain field.

According to the studies, AI models could assist dentists in predicting implant prognosis. AI could assist clinicians in treatment planning and evaluating prognosis for patients requiring dental implants [58]. Complications including peri-implantitis and marginal bone loss could be predicted in advance or detected at an early stage [51,54]. Therefore, implants with unclear prognosis will be recognized and treated prior, which will benefit both clinicians and patients. Meanwhile, osseointegration of implants could be partially predicted using DL models trained by radiographs [57]. Hence it would help dentists make rational clinical decisions after implant placement.

Despite the enormous advantages of AI for prediction, some limitations still exist. Firstly, radiographic data are mainly composed of 2-dimensional images. Periapical and panoramic radiographs are 2-dimensional images, which may not show the periapical lesions as clearly and accurately as CBCT [67]. Although CBCT was used in 2 studies [50,56], the images were sliced and trained by DL models, which will lead to bias during slicing. Huang et al. extract features from CBCT without dental implants before implant placement, lacking the post-operative X-rays or CBCT, which can provide information for clinicians and AI models to evaluate the peri-implant and bone conditions after surgeries [50]. Secondly, since implant loss is a complex multi-factorial event, more features extracted from clinical information are needed [50, 53,58]. Increased input features require a more complicated AI model to find out the relationship between features and output. Traditional ML algorithms may not perform well in complex situations with multiple input data [53]. Deep neural networks for classification will be suitable for the prognosis of dental implants with vast features [52]. Moreover, the size of the training, validation, and test datasets were limited [49,50, 53,54,58,60]. The performance may increase with a larger size of the dataset. Besides including more patients and implants, one solution to

increase the size of the dataset is data augmentation [50,60]. Data augmentation is a method to increase more samples by shearing, rotation, flipping, blurring, contrast stretching, and other operations without affecting image quality [68]. The operation may cause overfitting, requiring the researchers to find new types of data augmentation to increase the dataset [69]. Another method to obtain more samples is a web-based application by creating a website for users to apply the predictive model to their samples [54]. Users take advantage of the model and input their radiographs with labels at the same time, which will lead to a win-win situation. Furthermore, with the exception of Lyakhov et al., all data came from a single hospital or clinic [52]. The results will be influenced by the clinical skills of dentists and the specific features of patients in a certain region [50]. A multicenter retrospective and prospective study is required, which can increase the size of the dataset for training the AI models and decrease the influence of dentists and regional factors. Finally, all 12 studies establish segmentation and prognosis models, which can only be applied to predict whether the future of implants is satisfactory. Four of them lack follow-up after the process of treatment [49,55,57,59]. For future studies, adequate follow-up is required to evaluate the dental implants after a period of time. With sufficient follow-up data, a survival model can be established for prediction based on both outcome and time. Survival models can predict how much time a dental implant can survive or after how many months will a dental implant suffer from peri-implant diseases. This may provide clinicians with not only qualitative but also quantitative information on the prognosis of dental implants, assisting them in making a more rational clinical decision.

5. Conclusions

Within the limitations of this scoping review, it can be concluded that the performance of AI in the prognosis of implants partially met the demands of clinicians. According to the studies, the results including accuracy, sensitivity, specificity, and other evaluation criteria seem promising. More studies on the application of artificial intelligence in the prognosis of implants based on three-dimensional radiographic images, adequate clinical history, and clinical examination information are required for better prediction results.

CRedit authorship contribution statement

Ziang Wu: Writing – original draft, Visualization, Validation, Software, Investigation, Formal analysis, Data curation. **Xinbo Yu:** Writing – review & editing, Validation, Investigation, Data curation. **Feng Wang:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Chun Xu:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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